MOVENRUN · VERSION 3 · REVISION 2 · 31 AUGUST 2026

Seal it or lose it

The complete guide to how the game plays, how the money works, and the limit on every single thing in it. Written so that someone who has never touched crypto can read it end to end.

The decided design   Supersedes all six earlier documents   Base · one token · $MOVE

1 billion	2.00%	12 / 20	25%	0
HARD SUPPLY CAP NOBODY RAISES IT	LAND TOLL, ONCE SPLIT ACROSS EVERY DEED	CEILING PER SESSION AND PER DAY	THE DAILY RECHARGE THE ONE BILL	TOKENS CREATED BY LAND, STAKING OR PURCHASES

MovenRun is a development-stage product. This document describes intended design and is forward-looking. It is not financial, investment, legal or tax advice, it is not an offer, and it is not a promise of returns. $MOVE does not trade: there has been no sale, no listing, no liquidity and no allocation. Zone deeds are in-game digital assets and confer no real-world property rights. Every figure marked as a hypothesis is a modelled starting point to be replaced by measurement. Part Five states exactly which parts of this document are built and which are not.

CONTENTS

What is in this document

BEFORE YOU START

Three ways to read this

You want to play

Read Part One and stop. You will know everything you need and never have to think about the token at all.

You are deciding whether to back it

Read section 1, then Part Two, then sections 29 and 31. Section 31 is the honest account of what is actually built.

You are going to build it

Read all of it, then the full technical specification at `docs/GAME_ECONOMY_V3.md`, which carries the contract invariants and the settlement algorithm.

WHAT THIS DOCUMENT SETTLES

Six earlier documents disagreed with one another – the territory economy draft, the 25 August investment brief, the 27 August master design, the 30 August integrity review, the public docs site, and the code audit. This one decides every conflict between them and says which way and why. Where this document and an older one disagree, **this one is right and the older one is out of date.**

PART ONE

The game

What you actually do, how ground becomes yours, what it costs to keep, and every system that makes the next session worth going out for.

1

The whole thing on one page

You walk, run or cycle. The app proves the movement was real. If you get back to where you started, the ground you covered becomes yours on the map. Hold good ground long enough and it becomes a deed – a permanent record in your name that nobody can edit or take away. When other people move across it, a slice of their reward comes to you.

Move
Walk, run or ride. Ten minutes and 750 metres is enough to count. The app checks it was real movement, not a car and not a shaken phone.

Seal
Get back to your start, or to ground you already own, or cross your own trail. Until you do, you have claimed nothing.

Claim
Cells you stepped on become *solid*. Cells your loop enclosed but you never entered become *shade* – weaker, and anyone can take them by walking there.

Hold
Ground decays. Walking it again restores it for free. Not walking it costs a small daily upkeep, and the more you hold the more each zone costs.

Level
Spend what you earn to raise a zone from Outpost to Citadel, and your kit from Starter to Veteran. Levels buy defence and tools. They never buy income.

Own
Ground that real people cross, held long enough, becomes a deed. From then on, everyone who runs across it pays you a share of their reward.

The twelve rules that never change

1	Only **verified movement** creates $MOVE. Not owning land, not staking, not spending, not inviting friends.
2	There will **never be more than one billion** $MOVE. It is written into the contract with no function to raise it.
3	**Nothing is claimed until you seal it.** Go out and come back.
4	The toll is **2% of a session, once, in total** — however many people's land you crossed.
5	Land pays its holder **out of a runner's own reward**, never out of new supply.
6	There is a **ceiling on one session and a ceiling on one day**, and the day's ceiling is smaller than two sessions' worth.
7	Rewards **settle once a day**, shared out among the people who actually moved that day.
8	**Nothing you buy earns you anything.** Purchases buy defence, durability, identity and tools.
9	There is **one recurring bill** — the daily recharge — and it is always a share of what you earned that day. Never a flat price. Never payable with cash.
10	**No cost may ever exceed the reward that created it.** Every core loop completes without topping up.
11	Free players' rewards are **never sellable**, and never become sellable retroactively.
12	**Your route never goes on the blockchain.** Not raw GPS, not lists of map cells, not ever.

THE IDEA UNDERNEATH ALL TWELVE

The game must be excellent before the token matters, and every mechanism must have a stated limit. If a future feature breaks one of the twelve rules above, the feature is wrong – not the rule.

2

A session, start to finish

Tap start and you are moving within two seconds. No setup, no wallet prompt, no jargon, no wallet at all unless you go looking for it. Your route draws behind you as you go, with distance, time and pace kept glanceable so your eyes stay on the street.

Walking, running and cycling all count. There is no minimum speed. Cycling is scored on its own separate map, because a bike covers ground so much faster than a person on foot that mixing them would make the map unfair to walkers.

What counts as a session

RULE	SETTING	WHAT IT MEANS FOR YOU
Minimum to count	`10 minutes and 750 m`	Below this it is a walk, not a play. Accessibility alternatives are published separately
Pauses	`30-minute window`	Stop for a coffee and carry on — it is still the same session. You cannot split one walk into ten to farm rewards
Sessions that earn tokens	`2 per day`	A morning and an evening. A third still takes ground, still builds your streak, still earns XP and materials — just no tokens
One at a time	`1 live session per device`	Overlapping sessions are rejected

What “verified” means

Because territory is a real asset, the game cannot take your movement on faith. Every session is checked for physical plausibility: sensible speed, sane GPS accuracy, continuous motion, and a movement pattern that matches walking or running rather than sitting in a vehicle. Your phone’s motion sensors are checked against the GPS trace, and both are checked against the map.

A session that passes becomes a **compact proof** – a short, tamper-evident summary that the network can act on without publishing where you went. That proof is what settles. Your route never does.

> STATED HONESTLY
>
> No location system is perfectly spoof-proof, and we do not claim otherwise. A determined attacker with real devices moving in real places can extract some value – but at that point they are playing the game. The goal is not perfection. **The goal is that cheating costs more than playing.**

3

Sealing: get back, or get nothing

While you move, your route draws behind you as a live trail. The trail is not yours yet. It is a bet.

Nothing is claimed until the route is sealed. This is the single biggest change from every earlier version of the design, and it is what turns a fitness tracker into a game: every session becomes a hand of cards you have to get home to bank. There are three ways to seal, and you can use whichever suits the route you are on.

1 · Come home
Finish within **150 metres** of where you started. The classic loop, and what the route planner optimises for.

2 · Come to ground
Finish on any cell you already hold. Your own land is a safe harbour – which means owning land literally pays for itself.

3 · Cut your own line
Cross your own live trail and the loop you just closed seals instantly, while you carry on running.

> THE SAFETY RULE THAT MAKES THIS WORK
>
> An unsealed run still earns **full tokens, full XP, materials, contracts and your streak.** It only fails to claim ground. Nobody is ever stranded, rushed into traffic, or out of pocket because they needed to close a loop to get paid.
>
> **Land is what you play well for. Tokens are what you show up for.** Keeping those two separate is what makes the whole design safe.

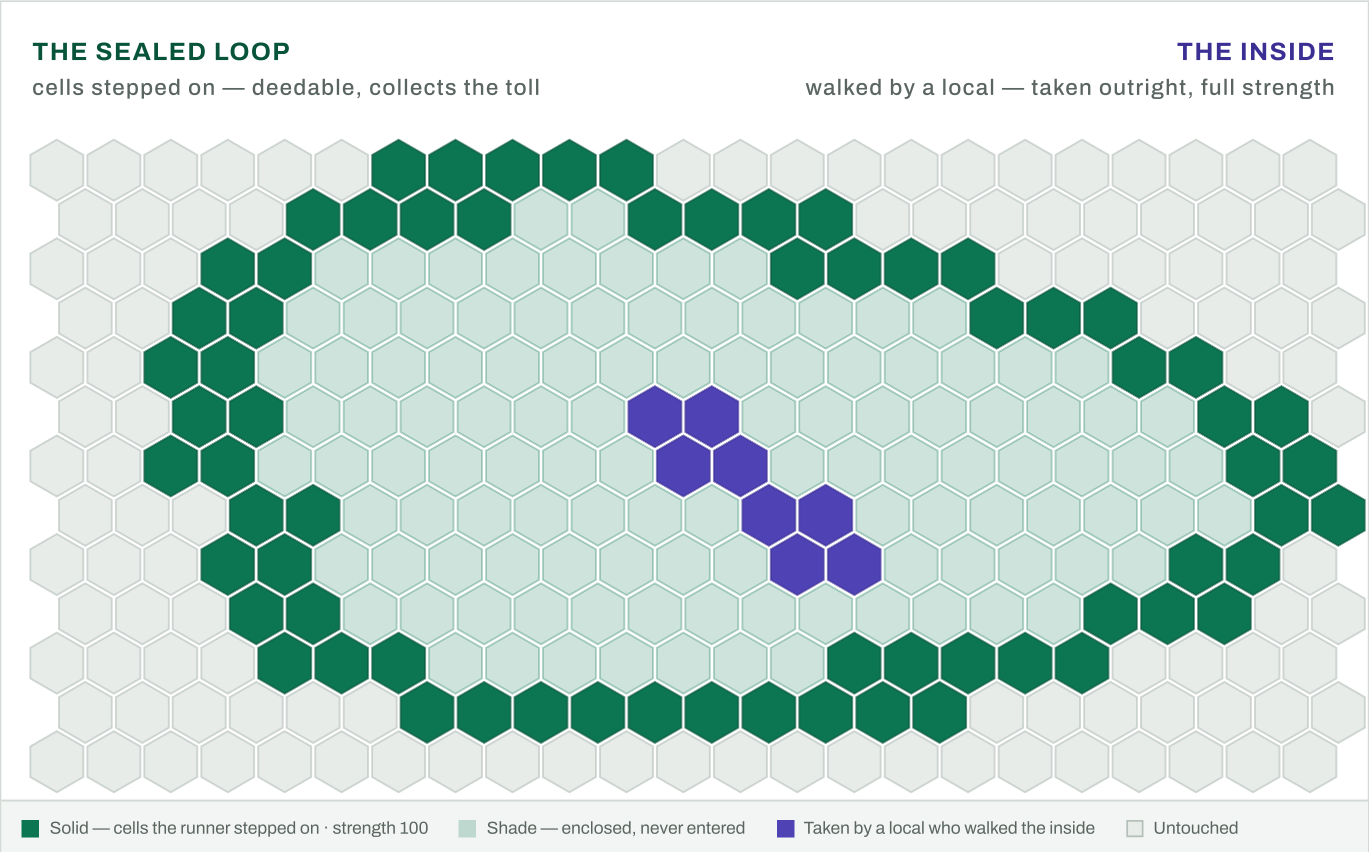
One optional mechanic, held back for later: if another player crosses your live trail before you seal it, your trail is cut and everything behind the cut is lost. It is the best player-versus-player idea available and it costs the token supply nothing – but it is also the most grief-prone thing in this document, so it does not ship until it has been simulated against an organised club.

4

Solid and shade

Sealing produces two very different kinds of territory, and only one of them is worth money. This is the answer to a question that three earlier documents left open: *can one person run around a whole city and own it?*

The answer is yes – and it will earn them nothing.



The ring runner holds the outline of the city and everything it encloses – on the map, on the leaderboard, and in the achievement. The walker who actually covered the streets inside holds the cells that can be deeded and that collect the toll.

	SOLID	SHADE
What it is	Cells you physically moved through	Cells your sealed loop enclosed but you never entered
Claim strength	`100`	`25 × shape × sprawl`
Decays	At the base rate	**Three times faster**
Can become a deed	**Yes**	**Never**
Collects the toll	**Yes**, once deeded	**Never**
Counts for map colour, area held, leaderboards, district control	Yes	Yes
How somebody takes it from you	By wearing it down, or by a formal contest once it is deeded	**Anyone who physically walks it takes it instantly, at full strength**

```
shape  = how round your loop was.  A circle scores about 1.
         A long thin sliver scores close to 0, so a boundary walk
         claims almost nothing in the middle.

sprawl = you may enclose up to 12 cells for every cell you actually
         stepped on at full strength.  Enclose more than that and the
         shade thins out in proportion.
```

SHADE IS MAP CONTROL. SOLID IS ECONOMY.

Run the entire ring road of a city and you get the city – in outline, in colour, on the leaderboard, with an achievement for it. Your shade over the centre will be worth a handful of strength points and will drain within days. Meanwhile the person who actually walks the high street takes those cells at full strength, keeps them, and is the one who can eventually deed them and collect the toll.

Both kinds of territory are real. Only one of them pays. That single distinction also closes a serious loophole: without it, one bicycle lap around a city's outer road would collect a toll from every runner inside it, for ever.

Never claimable, at any strength: military sites, schools, hospitals, shelters, private land and known hazards are masked out of the grid entirely.

5

Holding your ground

Every cell you hold has a **strength** from 0 to 100, and exactly one holder. Four things move that number.

WHAT HAPPENS	EFFECT
Somebody walks through it	Their push is subtracted from your strength — 100 for solid, the shade value for shade. When it hits zero the cell flips to them, carrying the overflow. Fortification blunts the incoming push
Time passes	Strength drains daily from the last time you were there. The stages are *held*, then *at risk*, then *contested*, then *dormant*, then back on the map — and every stage is on a calendar you can see in advance, never a surprise
You walk it again	**It resets to 100, instantly, for nothing.** Defence and exercise are the same action — that is the whole health idea in one line
You pay upkeep	Holds the line on a zone you did not visit. Covered in section 7

Deeded cells never wear down. They cannot be taken by someone walking through – only by a formal contest (section 13). They do still pay the toll to whoever crosses them.

6

The day-share: how every price works

One idea makes the whole economy immune to the network growing. It is worth ten minutes now, because every number in the rest of this document depends on it.

THE DEFINITION

A day-share is what the median active player earned that day. It is worked out at settlement and published. Every price in MovenRun – upkeep, the daily recharge, land levels, kit levels, claim fees, contest entry – is set in day-shares, and the app always shows you the figure in actual tokens before you commit.

Here is the problem it solves. A flat token price cannot survive the game succeeding. In season one with three thousand players, a good day pays close to the ceiling of 20 tokens. The same game with a hundred thousand players pays about 1 token a day, because the pool is fixed and more people share it (section 17). A fee of 4 tokens is a fifth of a day at the start, and four entire days later on. The same number would go from trivial to punishing without anybody changing it.

Priced in day-shares, **the same thing costs the same effort for ever** – at any population, at any token price, in any country. It also means we never have to run an emergency governance vote to re-price the game when it grows.

7

Upkeep and the skip penalty

If you visit your land, holding it is free. If you do not visit, you may pay to hold it instead. The charge is taken where it cannot fail to collect: **automatically, out of that day's reward, at settlement, before anything reaches your balance.**

```
upkeep owed = 0.03 day-shares × (zones held ^ 0.25)   for each zone you did not visit
```

The fourth-root term is what matters: the more you hold, the more *each individual zone* costs. Ten zones cost more each than one does.

ZONES HELD	COST PER ZONE/DAY	ALL UNVISITED, PER DAY	PER WEEK	WHAT THAT FEELS LIKE
1	0.030	0.03	0.2	Invisible
6	0.047	0.28	2.0	Comfortable for a committed local
16	0.060	1.00	7.0	**The break-even — one full day's earning, every day**
30	0.070	2.11	14.8	Needs real toll income to sustain
100	0.095	9.49	66	Needs a lot of it
250	0.119	29.8	209	Thirty days of earning, every day. Impossible

About sixteen zones is what movement income alone sustains if you never visit any of them. Visit them and the number is unlimited, because visiting is free. Own busy ground and the toll pays your upkeep for you – which is the design working as intended: *land that people actually cross pays for itself; land you hoarded and abandoned does not.*

> **THIS IS THE LIMIT THAT REPLACES AN ARBITRARY CAP**
>
> An earlier draft stopped large holders with a flat rule: no account earns more than 900 a day. That is a blunt instrument and it punishes genuine popularity. Upkeep does the same job honestly. Somebody who wants a whole city may have it – and will spend so much keeping it in condition that it stops being worth doing. **The limit emerges from the game instead of being imposed on it**, everything spent is destroyed, and a committed local holding six good zones never feels it.

You are never asked for more than you have

Upkeep is *owed* per zone, but what is actually *taken* is bounded. At settlement the system pays what it can out of that day's reward, up to a hard ceiling of 60% of it across all automatic charges combined. Anything still unpaid is **paid in condition instead** – those zones simply decay that day as though nothing had been paid on them.

So for somebody holding more than they can afford, the sequence is: pay what today's run covers, lose condition on the rest, watch the stages arrive on the calendar, and either go and walk them or let them go back to the map. **You never go into debt. Your balance never goes negative. No zone is ever lost without warning.**

Free grace, always: fourteen days of away protection per season, free, announced in advance, one tap. A holiday never costs you your territory, and you never have to buy protection. You pay only to be absent beyond that.

8

Charge and the daily recharge

There is exactly one recurring bill in MovenRun. This is it, and it is designed so that it can never become the thing that ruined the last generation of move-to-earn games.

How charge works

Your kit holds **100% charge**. Every session that earns tokens burns **50%** of it. Since two sessions a day is the most that can earn tokens anyway, a full day of play costs exactly one full charge. That is not a coincidence – it is what turns the two-session cap from an arbitrary rule into a resource you can see, plan around and spend deliberately.

Once a day, at settlement, your kit refills and the bill comes out of that day's reward.

```
recharge = charge used  ×  your kit's rate  ×  the day's reward

        100% used, Starter kit  →  25% of what you earned that day
         50% used, Starter kit  →  12.5%
          0% used               →  nothing at all
```

YOU RAN	CHARGE USED	STARTER KIT PAYS	VETERAN KIT PAYS
Nothing	0%	0	0
One scoring session	50%	12.5% of the day	7.5%
Two scoring sessions	100%	25% of the day	15%

Why a percentage rather than a flat number of tokens

You see a number of tokens – the app tells you the exact figure before you go out ("Today's recharge: 3.40 $MOVE") and again on the receipt afterwards. Behind that figure it is a percentage, for the reason set out in section 6: a flat price cannot survive the network growing. At a hundred thousand daily players the average day pays about 1 token, and a flat 4-token recharge would quietly have become a 400% tax on playing. A percentage costs the same effort at every scale, for ever.

The four rules that keep this safe

RULE	WHY IT EXISTS
Charge is spent on credit and settled at day close — you earn first, then the bill arrives	There is no state where you are too poor to recharge and therefore too uncharged to earn. **The deadlock cannot happen**
Running with an empty kit is always allowed	You still get territory, XP, streak, materials and contracts — everything except tokens for that session — and it costs no charge. **Charge gates minting, never play, and never your land**
There is no cash top-up. Ever	Recharge is paid in $MOVE from your own earned balance. There is no button that turns money into charge, because that button is exactly what made those other games predatory
All automatic day-close charges together are capped at 60% of the day's reward	Recharge plus upkeep can never take more than that, and can never push a balance negative. Past the cap, land condition degrades instead

Charge as a decision, not a chore

You can mark any session *unscored* before you start it. It costs no charge, earns no tokens, and still takes ground, still feeds your streak, still drops materials. So the short morning dog-walk can be free and both charges saved for the evening loop you actually care about – or spent early on a route you want to bank before somebody else runs it. That is a real decision, made several times a week, and it costs the economy nothing.

> **WHAT THE RECHARGE DOES FOR THE ECONOMY**
>
> At full participation this destroys roughly **a quarter of everything the game issues, every single day** – automatically, with no player decision required to fire it, and scaling with issuance rather than against it. It is the largest and most reliable sink in the design, and it is the main reason the supply can be expected to hold.

9

Levelling your land

Every zone you hold has a level. You raise it by spending what you have earned, plus time you have actually spent there. **Levels buy defence, capacity, identity and tools. Levels never buy income, and levels never buy a contest.**

LV	NAME	COST	ALSO REQUIRES	DECAY RESIST	EROSION RESIST	UPKEEP	AWAY DAYS	FORT SLOTS	WHAT IT UNLOCKS
1	**Outpost**	–	Claiming it	–	–	1.00×	–	1	Name your zone
2	**Camp**	2 day-shares	3 visits in 14 days	+8%	+5%	1.00×	+2	1	Banner, colour, zone photo
3	**Holding**	6	14 days held	+16%	+12%	0.95×	+4	2	**Rally beacon**, +10% materials, zone chat
4	**Stronghold**	18	30 days held, deeded	+24%	+20%	0.90×	+7	3	Sponsor tools, footfall analytics, leasing
5	**Citadel**	45	90 days held, survived a contest	+30%	+28%	0.85×	+10	3 + 1 permanent	Landmark status, icon on the city map, district vote weight

The five caps that keep levels honest

Decay resistance from level and fortification together	**+50% maximum.** A Citadel with three fortifications would be +60% on paper; it is capped. Land that survives a fortnight away is the product. Land that survives for ever is not
Erosion resistance	**+40% maximum.** A Citadel is hard to walk through. It is never impossible
Contribution to a contest	**Exactly zero.** The +15% ceiling on what money can add to a contest score is fortification only. Levels add nothing to it, at any level, at any price. The pay-to-win line does not move

Effect on toll income	**1.00× at every level.** A Citadel on a dead-end street earns less than an Outpost on the high street, for ever. Location is the only thing that makes land valuable
Losing the zone	It **drops two levels.** Reclaim it within 14 days and it is restored in full, with a Reconquest badge. That stake is what makes an upgrade a decision instead of a treadmill

10

Levelling your kit

The kit is the player-side tree, and what it buys is the most durable reward in the game: **a permanent reduction in your only recurring cost.**

LV	NAME	COST	RECHARGE RATE	ALSO GIVES YOU
1	**Starter**	–	25%	—
2	**Broken-in**	3 day-shares	23%	+10% materials
3	**Trail**	9	21%	Unused charge carries over to tomorrow, up to 50%
4	**Distance**	22	18%	One day off every cooldown; the full route planner
5	**Veteran**	50	15%	Ghost replays, +2 away days, a veteran mark on the map

Kit level never raises charge capacity. A third scoring session is not for sale at any price, to anybody. Kit level also never touches token earnings, territory strength, or a contest.

Fully levelling one zone is about 71 days of earning; fully levelling the kit is about 84. Those are long arcs on purpose – they are what a committed player works toward across a season and a half – and **every coin spent on either is destroyed.** All upgrade prices are calibration hypotheses, quoted in day-shares so they cost the same effort at any population.

11

Fortifying

Fortification is the short-term, expiring layer that sits on top of a level. It gives **+30% decay resistance and 30% less incoming erosion** while it lasts, up to three active at once, and it is burned whether you win or lose.

In a formal contest it contributes **at most +15%**, and that is the entirety of what money can do to a contest in this game.

> **A RISK WE ARE TAKING WITH OPEN EYES**
>
> The independent integrity review recommended that paid fortification have *no* effect on a contest at all. We have chosen a capped effect instead, because "pay to protect your land" is a mechanic players understand and it is a genuine sink. The +15% cap is what keeps it from becoming pay-to-win – and it **must be simulated against solo, small-club and large-club attackers before launch. If the simulation shows that it decides contests, the cap comes down or goes to zero.**

12

Deeds: permanent ownership

A deed is the permanent version of a claim: ground that a casual pass can no longer take, that collects the toll, and that you can sell. **You cannot buy one.**

An earlier design said "pay 400 tokens and your earnings become real". Read plainly, that is an investment product with a game around it, and it is the single thing most likely to fail an app store's or a regulator's reading of the product. It is gone. **Eligibility is earned, and the only currency is verified presence over time.**

REQUIREMENT	SETTING	WHY
Tenure	21 consecutive days	A deed upgrades ground you already hold, never ground you have never visited
Diverse organic traffic	30 distinct verified people, across 10 different days	Ground nobody crosses cannot be deeded. There is nothing to collect
Concentration limit	No related cluster above 20%	You cannot manufacture eligibility with a group of friends
Your own traffic	Never counts	Nor your friends', nor any other account of yours. This is checked with a risk graph, not on trust

REQUIREMENT	SETTING	WHY
Solid ground only	`Shade can never be deeded`	Section 4
Per-city cap	`Published share`	No person or club can corner one city's deeds
Claim fee	`Scales with how busy the ground is, burned`	It buys permanence, never income

The four layers of land

1 · Common ground
Free players capture and hold quiet areas. XP and locked rewards. No cost, not tradable.

2 · Zone badges
Non-transferable proof of local dominance, streaks and reconquests. Status, not an asset.

3 · Zone deeds
Permanent and transferable. Collects the toll. Can only be lost through a contest.

4 · Sponsored zones
Brands fund challenges around real locations. This is where outside money enters the game.

13

Contests: how a deed changes hands

A contest is declared, runs for a set window, and is settled by movement. It is the only way a deed moves without its owner selling it.

STAGE	RULE	WHY
Declared	A disclosed fee, burned. The deed goes into escrow	Stops spam. Buys entry, never score
Notice	72 hours before scoring even starts	You are never surprised out of something you hold
Window	7 days, and only your best 3 days count	Rewards turning up without demanding every single day. Rest days, work, illness and religious days cost you nothing
Score	Your verified movement + capped club support + a fixed 5–10% home advantage + fortification, maximum +15%	Nothing else is in the formula. If you cannot see it on that line, it does not affect the result
Club support	One support budget per person per contest, with diminishing returns	Thirty people cannot sweep a city
Cooldowns	A beaten attacker is locked out of that deed for 30 days; a survivor gets 14 days of peace	Stops anybody being griefed into quitting
Settlement	From escrow, and **anybody** can trigger it	The loser cannot refuse, stall, or move the deed. This was a real defect in our first contracts

The old design gave the current holder a +75% loyalty advantage, which made early owners effectively untouchable. That is gone, replaced by a fixed 5–10%.

14

Eleven things that keep it interesting

This is the gamification layer. **Not one item below creates a single token.** Everything here either costs nothing, moves tokens sideways between players, or burns them.

#	MECHANIC	HOW IT WORKS	WHY IT EARNS ITS PLACE
1	**The seal**	Your trail is a bet until you close it	The core tension. In every session, free, and it is the whole game in one rule
2	**Charge tactics**	Two charges a day; mark a session *unscored* to save one	Turns a cap into a decision. The morning walk becomes free; the evening loop becomes a choice
3	**Rally beacon**	Land level 3 and up. Light your zone for 24 hours and anybody who crosses it gets a contract tick and a materials bonus. Costs $MOVE, burned	**The only legitimate way to raise your toll income is to make people want to run there.** Landowners become promoters of their own neighbourhood, which is exactly the incentive a growing network needs
4	**Bounties**	Pin your own $MOVE to any zone: "first three verified crossers this week split it". Disclosed, skill-based, player-funded	Players write their own missions. Zero issuance — the tokens move sideways. It makes a quiet suburb interesting for a week
5	**The pincer**	Two people moving at the same time in the same city can seal the area *between* their two trails, and split it	Physically social. The best reason to bring a friend, and it costs the supply nothing

#	MECHANIC	HOW IT WORKS	WHY IT EARNS ITS PLACE
6	**Ghost of the holder**	Kit level 5. Entering a contested zone, you race the previous holder's route as a ghost	Turns an abstract contest into a head-to-head you can feel
7	**Streak shields**	Earned by hitting your own weekly goal. Spend one to protect a missed day. **Never purchasable**	Removes streak anxiety without removing streaks — and being un-buyable is what keeps it honest
8	**The Atlas**	A permanent, non-tradable record of every district you have ever sealed: the date, the shape of the loop, who held it before you	Pure status, and genuinely beautiful. The thing a five-year player has that a new one cannot buy at any price
9	**Season pass — free track only**	A 90-day objective ladder paying cosmetics, materials, shields and upgrade discounts. **There is no paid track**	All of the structure, none of the shakedown
10	**Reconquest**	Retake a lost zone within 14 days and its level is restored in full, plus a badge	Losing becomes a story instead of a wall
11	**Terrain and time**	Hills, parks, waterfront, night and rain change the *materials* drop table — never the token reward	Real reasons to vary your route, with zero effect on issuance

15

Clubs, districts and seasons

Alone you hold a neighbourhood. Together you take a city. Most of the long-term retention lives here, and none of it requires a token.

Every day

Three contracts – seal a loop over 1 km², cross two deeds, reconquer a lost cell. A **live pool ticker** showing today's pool, how many people are out right now, and your projected share. The city map redrawn at settlement: the daily who-took-what moment.

Every week

You set your own goal of 2 to 7 active days, and hitting *your* target pays a bonus. Crew missions. Club-versus-club district pushes.

Every season

90 days. District and city wars. Leaderboards layered block, suburb, city, country and world, so nearly everyone competes somewhere they can actually win. Landmark cells auctioned, carrying district vote weight.

Permanent, non-tradable badges: *City Outline* for running a whole city's boundary, *Reconquest*, *District Mastery* for the neighbourhoods you genuinely frequent, and *Cartographer* for the largest sealed loop of the season.

THREE THINGS DELIBERATELY NOT SHIPPING

Area-weighted prize draws. Every km² held being an entry in a monthly draw rewards players without printing anything, which is attractive – but a chance-based reward tied to holdings is a gambling-shaped mechanic with real app-store and legal exposure. Fixed, published sponsor rewards instead. If any draw mechanic ever ships it must be detached from any payment, publish its odds, and clear legal review in each market first.

A paid season track, a cash top-up for charge, and purchasable streak protection. These are the three levers that turn a fitness game into a shakedown, and none of them is worth what it would earn.

PART TWO

The money

Where tokens come from, the three ceilings on them, how a day is settled, how land earns, and where every token eventually goes to die.

16

What a token actually is

Forget everything you have heard about crypto for two minutes. The idea underneath is far simpler than the noise around it.

Start with a notebook that nobody owns. Imagine a notebook in a village square. Anyone can write in it, everyone can read it, and once a line is written it can never be erased – only added to. Thousands of people keep identical copies, so if one person forges their copy, every other copy disagrees and the forgery is thrown out.

That is a blockchain. Not a currency and not a company – just a way of keeping a record that no single party controls, *including the party that started it.*

A token is simply a balance in that notebook: "this account holds 350 units." What makes it interesting is not the technology but the rules attached to it – how many can exist, who may create them, what destroys them – which are written into the same notebook and cannot be quietly changed later. Those rules are what "tokenomics" means, and they are the subject of this part of the document.

> **IN ONE SENTENCE**
> A token is a balance in a ledger nobody owns, governed by rules nobody can secretly change.

Two kinds of money, kept carefully apart

Nearly every misunderstanding about a product like this comes from mixing up two separate systems. Once you see them apart, the rest is easy.

System one — the game's money
$MOVE. Players earn it by moving, spend it inside the game, and spending it destroys it. Almost no cash enters or leaves. **This is the game's fuel, not our income.**

System two — the business
Ordinary money. Sponsors back zones. Clubs and organisers subscribe. Cities fund health programmes. Events and analytics. **This is our actual revenue.**

The move-to-earn apps that collapsed had only system one. Their business *was* the token, so when the token fell they had no income and the product died with it. MovenRun is built so that **if $MOVE were worth nothing at all, the business would still have paying customers and the game would still be worth playing.**

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One tap, three ceilings

There is exactly one source of new $MOVE in the entire system: verified movement, drawing from a pool that is fixed in advance and does not care how many people play.

The pool

WHAT	SETTING	MEANING
Total supply	`1,000,000,000`	Written into the contract. No function to raise it, no admin override, no upgrade path that reaches it
Player rewards	`40% = 400 million`	The only pool that verified movement draws from
A season	`90 days`	Season one releases 9 million. Each season after is 10% smaller, down to a permanent floor of 2 million
How long that lasts	`About 44 years`	400 million over roughly 180 seasons. The schedule is generated from the allocation, so the two can never drift apart
The daily pool, season one	`100,000 $MOVE`	**The only tap in the system.** Everything else moves tokens sideways or destroys them

Burned tokens are destroyed permanently. They never restore room to issue more.

How your share is worked out

```
effort        = the square root of (your minutes ÷ 40), floored at 0.5, capped at 1.0
day points    = your two best qualifying sessions  ×  your consistency multiplier
the rate      = the day's pool  ÷  everybody's day points added together
your reward   = the rate  ×  your day points, then trimmed by the two ceilings
```

PART OF THE SCORE	RULE	REASON
Qualifying	`10 minutes, 750 m`	Below this, a session does not score at all
The effort curve	`Saturates at 40 minutes`	A 10-minute walk scores 0.5. A 40-minute run scores 1.0. **A three-hour run also scores 1.0**
Sessions that score	`2 per day`	Morning and evening. A third feeds the map, XP and streak only
Consistency	`×1.00 rising to ×1.50 at six active days`	**The largest lever in the whole formula.** A seventh day earns no more — rest is never punished
Trust weight	`0 to 1`	New, unverified or high-risk accounts score at reduced weight, and the difference stays in the pool rather than going to anybody else

Two 25-minute walks a day, six days a week, scores 2.37. One three-hour run once a week scores 1.00. **Consistency wins by well over a factor of two** – which is the health outcome we want – but the runner is never told that their run was worth literally nothing, which was the flaw in the previous version of this rule.

The two ceilings, and which one actually bites

CEILING	VALUE	THE JOB IT DOES
Per session	`12 $MOVE`	No single session can max out a day, which quietly pushes people toward two shorter sessions — the better health outcome
Per day, per person	`20 $MOVE`	The hard per-account cap. The ratio between the two is fixed at 0.6, not a coincidence
The daily pool	`100,000`	The system-wide cap. Absolute, always, regardless of anything else

Below about 5,000 daily players

A pro-rata share would exceed the daily cap, so **the ceilings bite** and the surplus rolls back into the season pool. This is the pilot. Without the ceilings, the first hundred players would each have been paid a thousand tokens a day.

Above about 5,000 daily players

The share falls below the ceiling and **the pool bites** instead. At a hundred thousand daily players the average day pays about 1 token. That is intended, and it has to be said plainly: token income per session shrinks as the network grows.

Whatever gets trimmed

Never goes to other players that day. It returns to the season pool and spreads across the season's remaining days. It never crosses a season boundary, and it can never raise the 400 million allocation.

SOMETHING THAT HAS TO BE SAID OUT LOUD

A shared pool makes fake accounts *more* attractive, not less: a thousand of them take a thousand shares out of the total, and a per-account cap does nothing about it because each fake account simply earns up to its own cap. **Three things actually stop it, and none of them is the cap.** First, fake accounts earn *locked* tokens, which can never be sold and never unlock retroactively – an account that costs nothing to make is worth nothing to farm. Second, trust weight, backed by passkeys, device history and a risk graph across shared devices and synchronised routes. Third, every account has to physically move, in a real place, for real minutes, under sensor checks.

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A worked day: Meera's Tuesday

Everything in Part Two, applied to one real day, with every number shown. This is the clearest way to understand the economy, and it takes five minutes.

The setting. Season one. The daily pool is 100,000 $MOVE. That Tuesday, 40,000 people ran, and everybody's day points added together came to 60,000. So the rate for the day is 100,000 ÷ 60,000 = **1.667 $MOVE per point**. The day-share – what the median active player earned – comes out at **2.50 $MOVE**, and is published.

What Meera did. A 25-minute walk before work and a 25-minute walk after dinner. It is her fifth active day this week. She holds six zones, all Camp level, and her route today happened to pass through two of them. She has a Starter kit and no deed yet. Her evening loop crossed three cells that other people hold deeds on: two belong to Ravi, one to Sam.

STEP	WHAT HAPPENS	WORKING	RESULT
1	Effort, each session	√(25 ÷ 40)	0.79 each
2	Both sessions	0.79 + 0.79	1.58 points
3	Consistency, 5 active days	1.58 × 1.40	2.21 day points
4	Her share of the pool	1.667 × 2.21	**3.68 $MOVE gross**
5	Ceilings checked	1.84 per session vs 12; 3.68 vs 20	Both clear. Nothing trimmed
6	The toll	3 deeded cells of the 12 she crossed, so 25% coverage: 3.68 × 2% × 0.25	-0.02
7	Split between owners	Ravi 2 cells, Sam 1	Ravi 0.012, Sam 0.006
8	Daily recharge	Both sessions scored, so 100% charge used, Starter kit: 25% of 3.68	**-0.92**
9	Upkeep	6 held, 2 visited today, so 4 unvisited: 4 × 0.03 × $6^{0.25}$ = 0.19 day-shares	-0.47
10	The 60% ceiling checked	(0.92 + 0.47) ÷ 3.68 = 38%	Clear. Nothing degrades
11	Credited to Meera	3.68 - 0.02 - 0.92 - 0.47	**2.27 $MOVE, locked**

What was destroyed
1.39 $MOVE – the recharge and the upkeep. Both burned, permanently, out of the day's issuance.

What moved sideways
0.02 $MOVE to Ravi and Sam. Nothing was created to pay them. Every coin they received was subtracted from Meera in the same instant.

What she also got
One sealed loop, three new cells, her streak intact, two of three daily contracts ticked, and a handful of hill materials. None of it costs the token supply anything.

Notice what did *not* happen. She was never asked to top anything up. She was never shown a countdown. Her rest day tomorrow will cost her nothing. And if she had walked all six of her zones instead of two, her upkeep would have been zero.

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A second day: Dan, who stopped

Dan built up 40 zones over a season. He has not run in four days, and his fourteen away days for the season are already spent.

WHAT	WORKING	RESULT
He earned today	He did not go out	0
Recharge	No charge used	0
Upkeep owed	40 × 0.03 × $40^{0.25}$ = 3.02 day-shares	7.55 $MOVE owed
Upkeep actually taken	60% of nothing is nothing	**0 tokens — all of it paid in condition**

All forty zones lose a full day of decay with nothing paid against them. His Camp-level zones resist a little, so the weakest go *at risk* first, and his calendar already shows the exact date each one turns *dormant*.

Dan has three options, and they are all reasonable. He can go and walk them, which is free and restores them completely. He can run tomorrow and let a slice of that reward cover the arrears. Or he can let the ones he does not really use go back to the map and keep the six near his house that he actually walks.

> **THE LESSON IN ONE LINE**
> The game asks for movement, and only asks for money from people who will not move. Nobody goes into debt, nobody has their balance seized, and nobody loses a zone they did not watch slide for a week first.

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The toll, from both sides

Move across ground somebody holds a deed to, and **2% of what you earned that session goes to them.** Not 2% per owner. 2% for the session, however many owners you crossed.

```
the toll   = your session reward  ×  2%  ×  the share of your route on other people's deeds
each owner = the toll  ×  their cells  ÷  all the deeded cells you crossed
```

YOUR SESSION	YOU EARN	TOLL	YOU KEEP	EACH OWNER RECEIVES
No deeded ground on your route	20.00	0.00%	20.00	nothing
Four of six cells deeded	20.00	1.33%	19.73	0.27 shared
Every cell deeded, **four owners equally**	20.00	2.00%	19.60	0.10 each — **0.50% each**
Every cell deeded, **eight owners equally**	20.00	2.00%	19.60	0.05 each — 0.25% each
Every cell deeded, twenty owners	20.00	2.00%	19.60	the same 2% split twenty ways

The more land you cross, the more finely the same two percent divides. Crossing twenty owners costs you exactly the same as crossing one. There is no version of this where the toll reaches 5%, or 3%, or 2.01%.

The rules on top of the formula

- Your own land is free to cross, and so is any account linked to you. **You cannot pay yourself.**
- Ten laps of the same cell in one session pay **one** toll, not ten.
- Rounding remainders go back *to the runner*, so the system never takes more than it disclosed.
- **Shade never collects a toll.** Only deeded solid ground does.
- Level, tier, fortification and spending have **no effect whatsoever** on toll income. Only location does.

What a deed is actually worth — Ravi's high street

Ravi holds one deeded cell on a busy high street that about 140 people cross on a normal weekday. Each of them hands him a small slice of their own 2%, averaging around 0.008 $MOVE.

WHERE A DEED IS	CROSSERS PER DAY	RELATIVE EARNINGS
Dead-end suburb	2	1×
Quiet residential street	12	6×
Busy path or park route	45	22×
City high street	140	70×

Ravi's cell brings in about 1.12 $MOVE a day, roughly 0.45 day-shares – comfortably more than ten times the upkeep on his three zones. The identical deed in a dead-end suburb earns a seventieth of that and does not cover its own costs. **That comparison is the entire land economy.** There is no tier column in this table, because tier does not affect income.

> **UNCAPPED FOR ONE PERSON, HARD-CAPPED FOR EVERYBODY**
>
> There is **no daily ceiling** on what a deed holder receives. If a hundred thousand real, verified people genuinely cross your high street, you collect from a hundred thousand real sessions. Genuine traffic is never punished.
>
> And the system is still bounded, because the toll is a slice of a fixed pool. In season one the pool is 9 million, so **the total toll paid to every landowner on Earth cannot exceed 180,000 $MOVE that season.** Uncapped for the individual, hard-capped for the system. Both statements are true at once, and that is the whole design.
>
> **Nothing is created.** Every coin an owner receives was subtracted from a runner in the same transaction. Land issues exactly zero new $MOVE, and settlement stops automatically if the books do not balance.

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Locked and liquid

Everyone earns the same token. Whether you can sell it depends on how you got it – and that single line decides every case in the economy.

Locked $MOVE	Liquid $MOVE
What every player earns by moving. Most of the supply. ✓ Buys everything in the game ✗ Cannot be sent, sold or withdrawn ✗ Never converts, not even later **Real progress. Zero sell pressure.**	Tokens that already existed, changing hands. ✓ Everything locked can do ✓ Transferable and sellable ✓ Carries voting rights ✓ Buys and sells deeds on the market **Same token, same ledger, same cap.**

One balance in the app, one number, one token. Nobody has to understand or manage two currencies.

HOW YOU GOT IT	STATE	WHY
A session reward, as a free player	LOCKED	Newly created
A session reward, once you hold a deed	LIQUID	From that moment onward — see below
A toll paid to you for crossing your deed	LIQUID	Transferred from another player, never minted
A sponsor prize	LIQUID	The sponsor deposited tokens that already existed
Bought on the open market	LIQUID	It already existed

THE RULE THAT HOLDS THE WHOLE THING TOGETHER

When a free player earns their first deed, **everything they earned before that moment stays locked for ever.** Only rewards earned after they qualify are liquid.

This single rule closes the loophole that would otherwise sink the design: farm for months across free accounts, qualify one of them, cash out everything. It has to be enforced in the token contract itself, not in the backend.

Three guardrails on land unlocking liquidity

Land cannot be bought
21 days' tenure, 30 distinct verified crossers over 10 different days, a concentration limit, and your own traffic never counts. You could burn a million tokens and still not qualify.

The revenue valve
Liquid issuance in any period is capped at what the business actually earned. If demand exceeds that, the excess **mints locked** and the shortfall is published. Sellable supply can never outrun real revenue.

The pool is still the ceiling
Liquid is a *state* of an already-capped issuance. It is not a second stream, and it cannot raise the total by a single token.

An honest note. "Acquire land and your rewards become sellable" sits close to being an investment gate, and it is worth naming that rather than hiding it. The three guardrails above are what make it defensible, and it needs legal sign-off in each market before it ships. If counsel objects, the fallback is one line of configuration: deed holders' rewards stay locked, and staking becomes the only route to a sellable token.

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Where tokens go to die

Ten things worth buying, and **not one of them earns you anything.** That last part is what stops the economy eating itself.

THE MISTAKE THIS RULE EXISTS TO AVOID

In the products that collapsed, the biggest thing you could buy was an item or a creature that then *earned tokens of its own*. So the largest sink was really an investment in more inflation. Every purchase today bought more minting tomorrow, and the books looked balanced right up until they did not.

Nothing you can buy in MovenRun produces tokens. Gear wears out. Condition drains. Fortifications expire. Levels change what you can do, never what you earn.

SINK	HOW OFTEN	WHAT IT IS
The daily recharge	Every day you played	**The largest sink in the design.** 25% of the day's reward on a Starter kit, 15% on a Veteran, pro-rated to the charge you actually used. Automatic, needs no decision, and scales with issuance
Upkeep / the skip penalty	Daily	The volume sink for landholders. Free if you visit instead, so it only ever charges absentees. Rises with how much you hold
Land levels	Rare, large	2 to 45 day-shares per zone. Buys defence, capacity, identity and tools — never income, never a contest
Kit levels	Rare, large	3 to 50 day-shares. Buys a permanently cheaper recharge
Fortifying	Weekly-ish	Burned whether you win or lose
Rally beacons	Weekly-ish	Buys traffic to your zone, which is the only honest way to raise toll income
Deed claim fee	Rare, large	Scales with how busy the ground is. Buys permanence, never income
Contest entry	Rare	Buys your place, never your score
Gear crafting, cosmetics, identity	Regular	Materials are terrain-gated, so they are a reason to go somewhere new. Cosmetics are a pure sink with zero advantage
Landmark auctions and the season burn	Seasonal	Rare city cells auctioned, proceeds burned; plus a scheduled burn at season end. **A real burn** — the currently deployed contract transfers to treasury and calls that a burn, which must be rebuilt

THE RULE THAT KEEPS THE SINKS HONEST

No cost may ever exceed the reward that created it. The recharge is a fixed share of the day's reward. Every automatic day-close charge together is capped at 60% of it. Upkeep can never push a balance negative – it degrades land instead.

An earlier draft calculated that a typical player earns about 60 a season and would want about 71 worth of things, and described the gap as a comfortable nudge to buy a top-up. That is an engineered shortfall dressed up as design, and it is forbidden here. **Every core loop completes without ever topping up.** If revenue depends on a gap we designed into progression, we built the wrong business.

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How the business actually earns

Four revenue engines. Three of them work identically whether $MOVE is worth ten dollars or nothing.

ENGINE	WHAT IT IS	PREDICTABILITY	NEEDS THE TOKEN?
Sponsors and brands	Zone sponsorship, city campaigns, footfall analytics. They buy verified footfall past their door — distinct visitors, repeat rate, month-on-month change, aggregated and thresholded, never individual paths	Medium, contract-based	No
Subscriptions	Premium features, club and race-organiser tools	High, recurring	No
Events	Races, city events, charity runs	Lumpy, high value	No
Land economy	Claim fees, a marketplace cut, leasing	Activity-driven	Partly

What we deliberately never charge for: playing, moving, capturing, streaks, clubs and city competition – free for ever, because charging there would kill the density everything else depends on. Basic access: no paywall, no trial, no "premium required to earn". And getting started: wallets are free, network fees are covered, and nothing must be bought to claim a first zone.

A sponsor can only *add* rewards. They can never take any away, gate anything, or pick a winner.

PART THREE

How it runs

Your wallet, the network fees somebody has to pay, what actually goes on the blockchain, what happens to your location data, and why cheating loses.

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Your wallet, made for you

You sign up with an email address or a phone number and a passkey – the same thing that unlocks your phone. Behind that, **a wallet is created for you automatically.** There is no seed phrase to write down, no chain jargon, no wallet screen at all unless you go looking for one.

That wallet is the account every transaction in the game runs through: your rewards land in it, your spending comes out of it, and your deeds are held in it. If you ever want to take control of it yourself, you can export it. If you never want to think about it, you never have to.

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Gas, and why nobody can drain it

Writing to a blockchain costs a small network fee, called gas. **We pay it for you**, through a Base paymaster. That immediately raises a fair question: what stops somebody using our gas budget for their own benefit? Six controls, all of which have to hold together.

#	CONTROL	WHAT IT STOPS
1	A strict allowlist. Only named functions on MovenRun's own contracts are ever sponsored. Arbitrary calls, arbitrary destinations, token approvals and raw transfers never are	Using our gas budget to run somebody else's transactions
2	Every sponsored action needs a short-lived server signature tied to the account, the day, a one-time number, and *a verified session*	Bot swarms that never actually move
3	A brand-new account gets zero sponsorship until its first verified session settles	Farming the gas budget by mass sign-up
4	Per-account quotas — ten sponsored actions a day — plus a ceiling on each one	One account draining the whole day's budget
5	A global daily budget with a circuit breaker that stops sponsorship and pages the on-call engineer	A bug or an attack costing an unbounded amount of money
6	There is no refund path. Sponsorship pays the network directly and never credits a player's account. No contract refunds gas to anybody	**Turning sponsored gas into value. There is simply nothing to extract**

AND THE WAY OUT IS NEVER SPONSORED-ONLY

You can always pay your own network fee to transfer or withdraw. If sponsorship is ever paused – for a bug, an attack, or a budget overrun – withdrawals and transfers keep working. A pause that can trap somebody's assets is indistinguishable from a rug pull, so we do not build one.

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What is on the blockchain, and what is not

Every token earned and every token spent is recorded on Base and can be independently verified. What changes is *how* it gets there: not one blockchain transaction per tap, but every action committed into a daily settlement that anybody can audit.

At a hundred thousand daily players doing five economic actions each, one transaction per action would be half a million transactions a day – 182 million a year, all of which we pay for. Settling in daily batches with pull-based claims brings that to roughly five thousand a day. The hundredfold difference is the point.

THING	ON-CHAIN?	HOW
You finish a session and earn	Yes	Committed into that day's settlement; your balance is provable from it
Recharge, upkeep, levels, fortify, gear	Yes	The same settlement, with a receipt naming exactly what it paid for

THING	ON-CHAIN?	HOW
A toll moving from runner to owner	Yes	A matched debit and credit in the same settlement. They must sum to zero
A deed created or transferred	Yes, immediately	Its own transaction. Ownership is never batched
A contest settling	Yes, immediately	Its own transaction, out of escrow
XP, levels, streaks, badges, condition, materials	No	The game ledger. Paying network fees for a number nobody trades is waste, it would publish a permanent record of your activity, and an on-chain XP balance that gates access to earnings starts to look like an asset — a question we do not need to invite
Your route, your cells, your GPS	**Never**	See section 27

The off-chain half cannot create tokens. Only the settlement contract can, only against that day's budget, and only with an authorised, expiring, chain-bound signature. Every input to the day's calculation is published in aggregate, the algorithm is deterministic and versioned, and any player can verify their own inclusion. A separate reconciliation job recomputes the result independently and **halts settlement if the books do not balance.**

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Your location data

Nothing about an individual's path goes on the blockchain. Ever.

An earlier design published, on a permanent public ledger, the list of map cells each session crossed. Raw GPS was correctly kept out – but a public sequence of map cells *is* a route. Anybody could read where a person walks, when, and roughly where they set out from. That is fixed here, and it was the most serious problem in the earlier documents.

DATA	WHERE IT LIVES	HOW LONG	WHO CAN REACH IT
Raw GPS and sensors	Encrypted, short-lived store	24–72 hours	Verification workers only
Fraud signals	Risk store, minimised	Reviewed	The fraud team, with an audit log
Your game route	Private game ledger	You choose	You, and services you authorise
Sponsor analytics	Aggregated and thresholded	Campaign term	No individual path, no re-identification
The public blockchain	Day totals and amounts only	Permanent	Everyone — which is exactly why it never contains a route

Safety defaults. We never publish your live route, your start point, your regular schedule or your crossing history. Start and end areas are blurred and public activity is delayed. A private account earns exactly as much as a public one. Military sites, schools, shelters, medical facilities, private land and known hazards are masked or blocked. Nothing in the game rewards speed over safety, rushing a road crossing, trespassing, or being somewhere dangerous, and you never need to look at the screen while moving.

Consent for location, fraud checks, marketing, sponsor analytics and public sharing are separate, revocable choices, and withdrawing one is as easy as giving it.

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Cheating, and why it does not pay

THE ATTACK	WHY IT LOSES
Fake the GPS	The app is checked for tampering. Sensor data is signed as it is recorded and must agree with the GPS trace and with the map. Each session is usable once. Nothing is created without the verification service's signature
Drive it slowly	A speed limit alone is not enough — slow city driving passes one. Movement is classified by mode using cadence, acceleration, stop pattern and altitude. A vehicle earns no walking territory
Shake the phone	Steps without displacement earn nothing; GPS and motion sensors have to agree
Run 100 km for a hundred times the tokens	**There is nothing to farm.** The effort curve saturates at 40 minutes and the per-session ceiling is hard. This removes the single largest cheating incentive that every previous version of this design had
Split one walk into ten sessions	Pauses inside 30 minutes merge into one session; a new one needs 10 minutes and 750 m of fresh movement; and only two a day score
Dodge the recharge by never settling	Charge is spent on credit and settled at day close. An unsettled day carries the charge forward. You cannot bank the rewards and skip the bill

THE ATTACK	WHY IT LOSES
Make a thousand accounts	They earn **locked** tokens that can never be sold, at reduced trust weight, and every one of them has to physically move. Passkeys, device history and a risk graph across shared devices and synchronised routes tie them together
Two accounts paying each other tolls	This creates zero tokens — it only moves a slice of an already-fixed reward between two accounts. The real risk is faking how popular a piece of land looks, so related traffic is excluded from the metrics that decide deed eligibility
One lap of the ring road to toll a whole city	**Shade collects no toll and can never be deeded.** This is the specific exploit that shade exists to kill
Manufacture traffic to qualify land	Eligibility needs 30 distinct verified people across 10 different days; related accounts and the owner's own traffic never count
Buy the whole map	Upkeep rises per zone as you hold more, levels add no income, there is a per-city deed cap, and everything spent trying is burned
A club sweeping every deed	One support budget per person per contest, with diminishing contributions, plus per-city limits
Grief one owner until they quit	A 30-day lockout after a loss, every attempt costs a burned fee, and the defender gets 14 days of peace
Drain the gas budget	The six controls in section 25 — and nothing to extract even if you did get sponsored
Compromise the signing key	Hardware-held keys, a separate key per job, short-lived signatures, rate bounds, rotation, and a circuit breaker that pauses new settlement. **Withdrawals and transfers are never frozen**

> **THE PRINCIPLE**
>
> Anything that can be automated earns nothing sellable. Anything sellable requires being in a real place, over real time.

When the system is unsure about you

STATE	WHAT YOU SEE	WHAT HAPPENS TO THE REWARD
Confirmed	Your receipt, immediately	Settles normally
Pending	A clear reason and an expected review time	XP may credit; tokens and any owner toll stay reversible until final
Not eligible	A specific explanation and how to fix it	No reward for that session. Unrelated balances are never confiscated
Appeal	Submit evidence in the app; a person reviews it	No permanent ban ever results from one weak signal

PART FOUR

Every limit

You asked that everything have a limit. This is the register. If a mechanic is not on this list, it does not ship.

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The limits register

WHAT	THE LIMIT	TYPE
SUPPLY AND ISSUANCE		
Total supply	1,000,000,000	FIXED
Player reward allocation	400M over ~180 seasons (~44 years)	FIXED
Season pool	9M, -10% each season, floor 2M	FIXED
Daily pool, season one	100,000 $MOVE	FIXED
Per-session ceiling	12 $MOVE	HYPOTHESIS
Per-day ceiling, per person	20 $MOVE	HYPOTHESIS
Ratio between the two	0.6 — one session can never max a day	FIXED
Whatever a ceiling trims	Back to the season pool; never crosses a season	FIXED
Sellable (liquid) issuance	Capped at real business revenue	FIXED
Staking allocation	100M total; deferred to phase 4	FIXED
SESSIONS AND SCORING		
Sessions that score per day	2	FIXED
Consistency multiplier	×1.50 at six active days; a seventh adds nothing	FIXED
Effort curve	√(minutes÷40), floor 0.5, saturates at 40 min	HYPOTHESIS
Minimum session that counts	10 minutes, 750 m	HYPOTHESIS
Session merge window	30 minutes	HYPOTHESIS
Trust weight	0 to 1; weight withheld stays in the pool	HYPOTHESIS
CHARGE AND THE DAILY BILL		
Charge per scoring session	50% of 100	FIXED
Daily recharge price	25% of the day at Kit 1, 15% at Kit 5	HYPOTHESIS
All automatic day-close charges combined	60% of the day's reward, maximum	FIXED
Cash top-up for charge	Does not exist	FIXED
Running with an empty kit	Always allowed; earns everything but tokens	FIXED
Charge capacity	Not purchasable at any price	FIXED
TERRITORY		
Sealing	150 m, own ground, or crossing your own trail	HYPOTHESIS
Solid claim strength	100	FIXED
Shade claim strength	25 × shape × sprawl	HYPOTHESIS
Shade sprawl bound	12 enclosed cells per stepped cell	HYPOTHESIS
Shade decay	3× the solid rate	HYPOTHESIS
Shade: deeds and toll income	Never	FIXED
Deeded cells	Cannot be worn down; contest only	FIXED

WHAT	THE LIMIT	TYPE
Restricted areas	Unclaimable at any strength	FIXED
HOLDING, LEVELS AND DEFENCE		
Upkeep / skip penalty	0.03 day-shares × $zones^{0.25}$, per unvisited zone per day	HYPOTHESIS
Unpayable upkeep	Paid in condition. Balances never go negative	FIXED
Away protection	14 days a season, free	FIXED
Land levels	5, costing 2 to 45 day-shares, burned	HYPOTHESIS
Kit levels	5, costing 3 to 50 day-shares, burned	HYPOTHESIS
Decay resistance, level + fortify together	+50% maximum	FIXED
Erosion resistance, level + fortify together	+40% maximum	FIXED
What a level adds to a contest	Zero. The +15% is fortification only	FIXED
What a level adds to income	1.00× at every level	FIXED
Losing a zone	Drops two levels; full restore if reclaimed in 14 days	FIXED
Fortifications active at once	3 per zone, each expires	FIXED
Fortification in a contest	+15% maximum	FIXED
THE TOLL AND DEEDS		
Toll rate	2.00% of a session, once, in total	FIXED · NOT GOVERNABLE
How the toll splits	By deeded cells crossed, per owner	FIXED
Total toll to all owners, season one	180,000 $MOVE	STRUCTURAL
One owner's daily income	No cap	DELIBERATE
Repeated laps	Toll once per cell per session	FIXED
Deed tenure and traffic	21 days; 30 distinct people over 10 days	HYPOTHESIS
Deeds per person per city	Capped	FIXED
Contest: notice and window	72 hours, then 7 days, best 3 count	FIXED
Home advantage	5–10%	HYPOTHESIS
Attacker cooldown / defender peace	30 days / 14 days	FIXED
INFRASTRUCTURE, PRIVACY AND GOVERNANCE		
Sponsored actions per account per day	10, gated on having played	HYPOTHESIS
Sponsorship for a brand-new account	Zero until the first verified session	FIXED
Raw location retention	24–72 hours	TARGET
Route data on the public chain	None, ever	FIXED
Governance over the toll or the supply cap	Impossible	FIXED
What a pause can cover	New claims and settlement only. Never withdrawals	FIXED
Paid season track, purchasable streak shields	Do not exist	FIXED

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What we promise never to do

WE WILL NEVER	BECAUSE
Make a rest day feel like a loss	Weekly goals you set yourself, not daily streak punishment. The consistency multiplier caps at six days for exactly this reason
Charge you to protect what you earned	Visiting is free, away protection is free. No fee to stop decay, no subscription to keep territory
Let you buy a contest win	Money buys entry, durability and appearance. Movement decides outcomes, and fortification is capped at +15%

WE WILL NEVER	BECAUSE
Design a gap and then sell you the fix	Every core loop completes without a purchase, and no cost may exceed the reward that created it
Sell charge for cash	That single button is what turned the last generation of these games into a treadmill
Use fake urgency	No countdowns unrelated to real events, no loss-framed notifications, no confetti over a money figure
Pre-select a purchase or auto-top-up	Exact cost, exact benefit, no guaranteed return — shown before you commit, every time
Promise income, a payback period or a rate	No yield quoted, no APR, no price
Freeze your exit	If we pause anything we pause new claims — never withdrawals or transfers. A switch that can trap your assets is indistinguishable from a rug pull
Change the rules retroactively	Every session records the rules version it was played under
Quietly seize an account	Recovery has a visible delay, notifies every channel, and an advanced user can opt out of it entirely

Everything is switchable. Notifications, public leaderboards, the token price display, social comparison and wallet export can each be turned off independently. **Somebody who wants a quiet walking app should be able to have exactly that.**

PART FIVE

Reality

What is actually built today, what is not, the order it gets built in, and the vocabulary in one place.

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Where the code actually is

READ THIS BEFORE SHOWING ANYONE THE REST OF THIS DOCUMENT

None of the economy in Parts One and Two is implemented. Eight contracts exist on Base Sepolia – a test network – implementing an older, different design, and a self-audit documents sixteen defects in them, three critical. **No deed has ever been created on any network. No token has ever reached a player. There is no deployment on Base mainnet.**

COMPONENT	STATUS	THE REALITY
Mobile app — sessions, GPS, map, progression, decay, defend, fortify	BUILT	Real and tested. The decay and fortify logic already exists locally
Movement verification pipeline	BUILT	Validates and produces a signed proof — but nothing submits it to a chain
Accounts, sessions, identity, wallet abstraction	BUILT	The strongest code in the project
Contracts on Base Sepolia	OLDER DESIGN	Eight deployed, sixteen documented defects, three critical
Real hex indexing in the app	NOT BUILT	**The app uses a local ~300 m lattice, explicitly marked "not real H3", while the contracts key deeds by H3 cell id. These index different worlds and must be reconciled before any integration.** No amount of economy design fixes this one
Deed eligibility from real traffic	NOT BUILT	The traffic function returns hard-coded zeros, so no deed can be created today by anyone
The 2% toll	NOT BUILT	The word "toll" appears nowhere in the codebase. The deployed contract has a mint-time zone tax, which is a different mechanic
Sealing, shade, erosion	NOT BUILT	New in this document
Daily settlement, the ceilings, trust weight	NOT BUILT	New in this document. The deployed contract pays per kilometre
Charge, the daily recharge, land and kit levels	NOT BUILT	New in this revision
Upkeep scaling with holdings	NOT BUILT	Decay exists locally; the scaling curve does not
Contests with notice and escrow	NOT BUILT	The deployed version cannot settle without the loser's cooperation
Embedded wallets, paymaster, sponsored fees	NOT BUILT	**Zero occurrences of "paymaster", "4337", "bundler" or "gasless" anywhere in any source file.** No provider selected. This is the largest unbuilt dependency in the design
Anything on Base mainnet	NOTHING	—

Three statements on the public site that must be corrected now

1. It says MovenRun "runs on Base" without saying testnet, and that deeds, balances and governance "live onchain". They do not yet.
2. It says privileged actions are "gated by explicit roles, not a single all-powerful key", and that the system is timelocked. Today **one key controls all eight deployed contracts and no timelock exists.**
3. It promises deed holders "in-game yield" and describes staking as live. This design removes passive yield entirely and defers staking.

None of these is fatal. All of them are fatal if an outside reviewer finds them before we fix them.

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The order we build it in

PHASE	SCOPE	WHAT MUST BE PROVEN BEFORE MOVING ON
0	**Simulation**	The daily budget holds. The toll invariants balance. The ceilings bite where section 17 says they do. Shade cannot be farmed. The recharge is affordable at every population. Upkeep scaling actually deters hoarding. Sessions cannot be split. Contests are fair to solo, small-club and large-club attackers **with fortification at +15% — and if not, it goes to zero**. Fake-account economics under a shared pool
1	**Credits-only pilot 1,000–5,000 players**	**The phase that matters.** Sealing, shade, erosion, levels, charge and daily settlement, with no transferable reward at all. **If the territory loop is not fun without money, money will not save it**
2	**Limited $MOVE small verified group**	Every day's books balance. Appeals work. Location deletion verifiably happens. Owner tolls provably come from runner rewards. The paymaster policy holds under attack
3	**Multiple cities 10,000–25,000**	Geography partitions cleanly. Club controls hold. Market manipulation tests pass. **Legal sign-off on land unlocking liquidity, before liquid earnings ship**
4	**Full scale, then staking**	Load tested at twice projected peak. External security audit complete. Incident drills run. App-store and legal sign-off obtained

> **THE NUMBERS TO BE IMPATIENT ABOUT**
>
> Week-4 retention above 25%. A thousand active players in one city. More than 20% of zones contested. Three sessions in an active week. Club participation above 30%. Three signed sponsors in city one. A stable ratio of verified to rejected routes.
>
> **Two matter more than all the others.** Week-4 retention says whether people form the habit. The first *renewed* sponsor says whether anyone will pay for the audience that habit creates. Every other metric is a detail beside those two.

Phase 1 is the one that decides everything. A territory game people love without any token is a business. A token attached to a game nobody loves is a liability.

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Glossary

Blockchain	A shared record that thousands of people keep identical copies of, so no single party can alter it — including us
Base	The blockchain MovenRun runs on. Cheap and fast enough that a game can afford to write to it
Burn	Permanently destroying tokens. Gone; nobody holds them afterwards, and the room they freed is never reissued
Charge	Your kit's fuel. 100 units; each scoring session uses 50. Refilled once a day, paid for out of that day's reward
Condition / strength	A zone's health, 100 down to 0. Neglect wears it; visiting or paying restores it
Consistency multiplier	×1.00 to ×1.50 based on how many days you were active this week. The biggest lever in the reward formula
Contest	The seven-day competition for somebody else's deed. Decided by movement
Day-share	What the median active player earned that day. Every price in the game is quoted in these, and displayed to you in tokens
Deed	The permanent record proving you hold a cell. A game asset, not real property
Erosion	Losing strength because somebody walked through your cell
Escrow	Holding an asset inside a contract so neither side can move it until a condition is met
Fortify	Paying to strengthen a zone temporarily. Capped at +15% in a contest
Gas	The small fee for writing to a blockchain. Covered by MovenRun
H3	The open standard that divides the world into hexagons. We did not invent it and cannot change it
Kit	What you carry. Its level sets how cheap your daily recharge is
Liquid $MOVE	Transferable and sellable. Comes from tolls, deed-holder rewards, sponsor prizes, or the market
Locked $MOVE	Spendable inside the game, never transferable. What movement rewards arrive as
Oracle	The signing service that tells the contracts a session really happened. It can sign; it can never move anyone's money

Paymaster	The thing that pays your network fees for you, under the six controls in section 25
Proof of movement	A short, tamper-evident summary of a verified session. Provable without revealing where you went
Season	90 days. Each releases a smaller pool than the last, down to a permanent floor
Seal	Closing your route so it claims ground — back to your start, onto your own land, or across your own trail
Settlement	The once-a-day on-chain commitment of every earn and every spend
Shade	Ground your loop enclosed but you never entered. Weak, fast-decaying, never deedable, never earns
Sink	Anything that removes tokens from circulation permanently
Solid	Ground you actually stepped on. Full strength, deedable, collects the toll
Toll	2% of a runner's session reward, shared among the owners of deeded ground they crossed. Never more than 2%
Trust weight	How much of the pool your account scores for, from 0 to 1, based on verification history
Upkeep	What it costs to keep a zone you did not visit. Rises as you hold more
Wallet	The thing that holds your $MOVE and your deeds. Created for you; you never have to see it

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The six sentences that hold the whole design

1. **One source.** Only verified movement creates $MOVE, from a fixed daily pool that does not care how many people play.
2. **Seal it or lose it.** Tokens are what you show up for; land is what you play well for, and nothing is claimed until you get back.
3. **Solid pays, shade does not.** You can control a whole city on the map by running its outline, and still earn nothing from ground you never stepped on.
4. **No second source.** Land pays its holder out of a runner's own reward. Two percent, once, split however many ways.
5. **Nothing bought ever earns.** Every ceiling is published, and the more land you hold the more it costs to keep.
6. **One bill, always affordable.** The daily recharge is a share of what you earned that day, it cannot be paid with cash, and an empty kit never stops you playing or costs you your land.

Every rule in this document is an application of one of those six. **If a future feature breaks one of them, the feature is wrong.**

> The full technical specification — settlement invariants for the fuzz test suite, the contract responsibilities, the exact on-chain and off-chain boundary, and the complete decision log against the six earlier documents — lives at `movenrun/docs/GAME_ECONOMY_V3.md`. Where this guide simplifies for readability, that document is authoritative.